

Marshallian vs. Hicksian Demand

ECO 310 Study Guide

1 Overview

There are two fundamental types of demand functions in consumer theory, each answering a different optimization problem.

Type	Problem	What's Fixed?
Marshallian	Maximize utility given income	Income m and prices
Hicksian	Minimize expenditure given utility	Utility level $\bar{u}$ and prices

2 Marshallian Demand

2.1 Definition

The **Marshallian demand function** $x^*(p, m)$ solves:

$$\max_x u(x) \quad \text{subject to} \quad p \cdot x \leq m$$

This answers: “What bundle should I buy to maximize my utility, given my income m and prices p ?”

2.2 Properties

- Depends on **prices** p and **income** m : $x^*(p, m)$
- Also called **uncompensated demand** (income is not adjusted for price changes)
- **Homogeneous of degree zero**: $x^*(\lambda p, \lambda m) = x^*(p, m)$ (no money illusion)
- Satisfies **Walras’ Law**: $p \cdot x^*(p, m) = m$ (consumer spends full budget)

2.3 First-Order Conditions

At the optimum:

$$\frac{MU_i}{MU_j} = \frac{p_i}{p_j} \quad \text{for all goods } i, j$$

Or equivalently: “bang per buck” is equalized across all goods:

$$\frac{MU_i}{p_i} = \frac{MU_j}{p_j}$$

2.4 Example: Cobb-Douglas

For $u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$:

Marshallian demands:

$$x_1^*(p_1, p_2, m) = \frac{\alpha m}{p_1}, \quad x_2^*(p_1, p_2, m) = \frac{(1-\alpha)m}{p_2}$$

The consumer spends a constant fraction α of income on good 1, and $(1-\alpha)$ on good 2.

3 Hicksian Demand

3.1 Definition

The **Hicksian demand function** $h(p, \bar{u})$ solves:

$$\min_x p \cdot x \quad \text{subject to} \quad u(x) \geq \bar{u}$$

This answers: “What’s the cheapest way to achieve utility level $\bar{u}$, given prices p ?”

3.2 Properties

- Depends on **prices** p and **target utility** $\bar{u}$: $h(p, \bar{u})$
- Also called **compensated demand** (utility is held constant)
- **Homogeneous of degree zero**: $h(\lambda p, \bar{u}) = h(p, \bar{u})$
- By construction: $u(h(p, \bar{u})) = \bar{u}$ (achieves exactly the target utility)

3.3 First-Order Conditions

At the optimum, the same tangency condition holds:

$$\frac{MU_i}{MU_j} = \frac{p_i}{p_j}$$

But now the constraint is $u(x) = \bar{u}$ instead of $p \cdot x = m$.

3.4 Example: Cobb-Douglas

For $u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$:

Hicksian demands:

$$h_1(p_1, p_2, \bar{u}) = \bar{u} \left(\frac{\alpha}{1-\alpha} \right)^{1-\alpha} p_1^{-\alpha} p_2^{\alpha-1}$$
$$h_2(p_1, p_2, \bar{u}) = \bar{u} \left(\frac{\alpha}{1-\alpha} \right)^{-\alpha} p_1^\alpha p_2^{-\alpha}$$

(These look messy, but the key is they depend on $\bar{u}$ instead of m .)

4 The Expenditure Function

The **expenditure function** $e(p, \bar{u})$ is the minimum expenditure needed to achieve utility $\bar{u}$ at prices p :

$$e(p, \bar{u}) = p \cdot h(p, \bar{u})$$

For Cobb-Douglas:

$$e(p_1, p_2, \bar{u}) = \bar{u} \cdot p_1^\alpha p_2^{1-\alpha} \left[\alpha^{-\alpha} (1-\alpha)^{-(1-\alpha)} \right]$$

5 Relationship Between Marshallian and Hicksian Demand

The two demand functions are **dual** to each other. They give the same answer at corresponding income and utility levels.

5.1 Key Relationship

If $x^* = x^*(p, m)$ is the Marshallian demand at income m , and achieves utility $u^* = u(x^*)$, then:

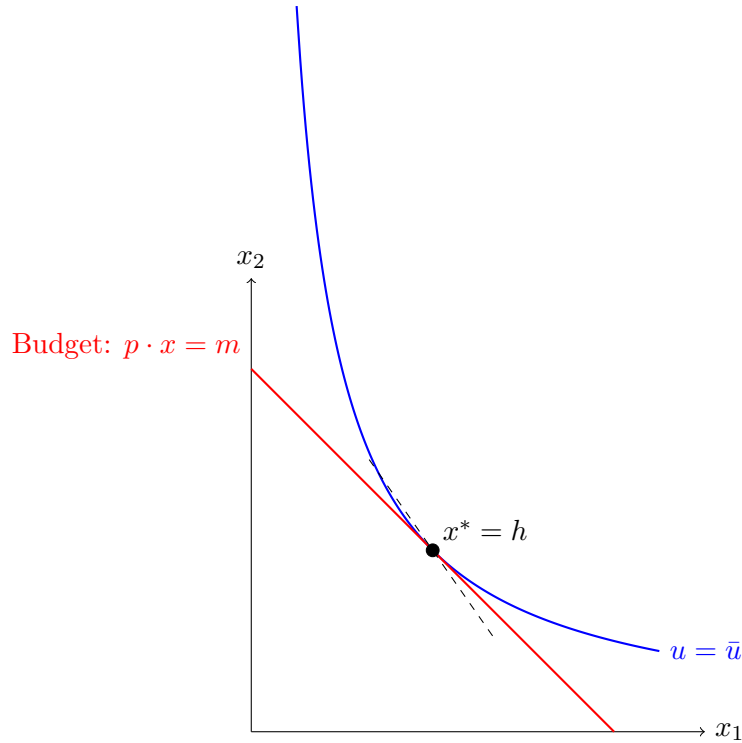
$$\boxed{x^*(p, m) = h(p, u^*) \quad \text{and} \quad h(p, \bar{u}) = x^*(p, e(p, \bar{u}))}$$

In words:

- The utility-maximizing bundle at income m equals the cost-minimizing bundle for the utility level you achieve
- The cost-minimizing bundle for utility $\bar{u}$ equals the utility-maximizing bundle at the expenditure level $e(p, \bar{u})$

5.2 Graphical Interpretation

Both problems give the same tangency point:



Both the Marshallian problem (maximize utility on the budget line) and the Hicksian problem (minimize cost to reach the indifference curve) give the same tangency point.

6 Income and Substitution Effects

The difference between Marshallian and Hicksian demand is crucial for understanding how demand responds to price changes.

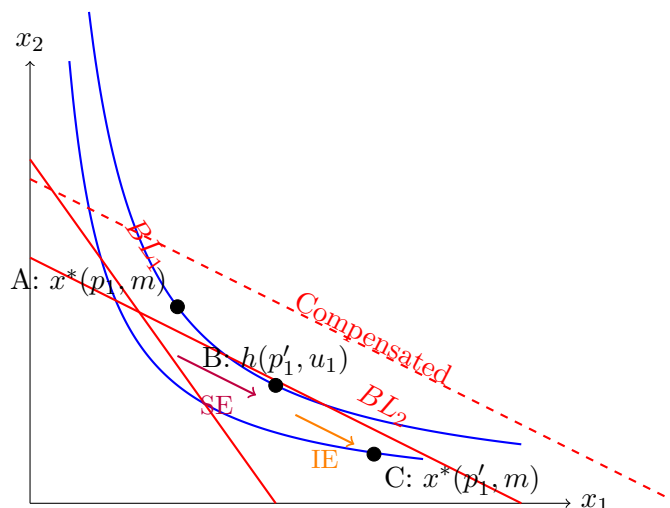
6.1 Slutsky Equation

When price p_i changes, the total change in Marshallian demand can be decomposed:

$$\frac{\partial x_i^*}{\partial p_i} = \underbrace{\frac{\partial h_i}{\partial p_i}}_{\text{Substitution Effect}} + \underbrace{-\frac{\partial x_i^*}{\partial m} \cdot x_i^*}_{\text{Income Effect}}$$

- **Substitution Effect:** Change in Hicksian demand (pure substitution, utility held constant)
 - Always negative: $\frac{\partial h_i}{\partial p_i} < 0$ (law of compensated demand)
- **Income Effect:** How the price change affects purchasing power, and how that affects demand
 - Sign depends on whether the good is normal ($\frac{\partial x_i^*}{\partial m} > 0$) or inferior ($\frac{\partial x_i^*}{\partial m} < 0$)

6.2 Graphical Decomposition



When p_1 decreases (budget line rotates from BL_1 to BL_2):

- **A to B:** Substitution effect (compensated) – always increases demand for good 1
- **B to C:** Income effect – depends on whether good is normal or inferior
- **A to C:** Total effect = SE + IE

7 Summary Table

Feature	Marshallian $x^*(p, m)$	Hicksian $h(p, \bar{u})$
Problem	Max u s.t. $p \cdot x \leq m$	Min $p \cdot x$ s.t. $u(x) \geq \bar{u}$
Arguments	Prices p and income m	Prices p and utility $\bar{u}$
Other name	Uncompensated demand	Compensated demand
Observable?	Yes (from market data)	No (requires utility info)
Price effect	Total = SE + IE	Pure substitution (SE only)
Always downward sloping?	No (Giffen goods)	Yes (law of compensated demand)

8 Key Takeaways

1. **Marshallian demand** is what we observe in real markets – how much people buy given their income and prices.
2. **Hicksian demand** is a theoretical construct that isolates pure substitution effects by holding utility constant.
3. The two are related through the **expenditure function**: $h(p, \bar{u}) = x^*(p, e(p, \bar{u}))$
4. The **Slutsky equation** decomposes price effects into substitution (Hicksian) and income components.
5. **Hicksian demand always slopes down** (compensated law of demand), but Marshallian demand can slope up for Giffen goods (when negative income effect dominates).